

Glomerular (Nephritic) — ANCA Vasculitis (GPA / MPA / EGPA)



1. ANCA small-vessel vasculitis

WHAT YOU SEE

A wooden banner reading ANCA SMALL-VESSEL VASCULITIS — THE TRIPLETS, presiding over a workshop with three worker tables — the three tables ARE the three sibling diseases GPA, MPA and EGPA.

WHAT IT ENCODES

ANCA-associated vasculitis is a pauci-immune SMALL-vessel vasculitis comprising three entities — GPA (granulomatosis with polyangiitis), MPA (microscopic polyangiitis), and EGPA (eosinophilic GPA / Churg-Strauss). ANCAs (antineutrophil cytoplasmic antibodies) activate primed neutrophils that degranulate and damage capillary/venule/arteriole walls, classically producing a pulmonary-renal syndrome. It is the most common cause of rapidly progressive (crescentic) glomerulonephritis in adults.

BOARD TRAP

Do not confuse small-vessel ANCA vasculitis with MEDIUM-vessel polyarteritis nodosa (PAN): PAN is ANCA-NEGATIVE, spares the lungs and glomeruli, causes microaneurysms ('rosary beads' on angiography), and is associated with hepatitis B — picking PAN for an alveolar-hemorrhage-plus-GN stem is wrong.

2. Pauci-immune

WHAT YOU SEE

A placard reading PAUCI-IMMUNE pinned above the workbenches — the near-empty staining stands for the few-to-no immune deposits seen on immunofluorescence.

WHAT IT ENCODES

Immunofluorescence shows FEW or NO immune deposits ('pauci-immune'), and electron microscopy shows scant deposits — the inflammation is antibody-driven against neutrophils, not from immune-complex or anti-GBM deposition. This is one of the three IF patterns of crescentic GN (the others being linear anti-GBM and granular immune-complex).

BOARD TRAP

The WRONG answer is LINEAR IgG along the GBM — that is anti-GBM (Goodpasture) disease, which also causes pulmonary-renal syndrome but is treated with urgent plasmapheresis + immunosuppression and is distinguished by anti-GBM antibodies. Granular ('lumpy-bumpy') IF instead indicates immune-complex GN (e.g., lupus, post-infectious).

3. Crescentic RPGN

WHAT YOU SEE

A microscopy panel of a glomerulus crammed with a crescent-shaped wedge of cells (tagged RENAL-LIMITED) — the cellular crescent filling Bowman's space IS the crescentic, rapidly progressive GN.

WHAT IT ENCODES

ANCA vasculitis causes crescentic, rapidly progressive glomerulonephritis (RPGN): biopsy shows cellular crescents (proliferating parietal epithelial cells, macrophages, fibrin) filling Bowman's space and focal segmental necrotizing GN. Clinically the nephritic picture is hematuria with dysmorphic RBCs and RBC casts, sub-nephrotic proteinuria, hypertension, and a creatinine that doubles over days to weeks. A renal-limited form (no extrarenal disease) exists.

BOARD TRAP

RPGN is a NEPHROLOGIC EMERGENCY — the trap is delaying diagnosis to 'observe,' allowing irreversible glomerular scarring. Days-to-weeks of rising creatinine with active urine sediment demands urgent biopsy and empiric immunosuppression, not watchful waiting.

4. Normal complement

WHAT YOU SEE

A small card on the induction wall reading COMPLEMENT NORMAL — the unchanged level signals this pauci-immune disease does NOT consume complement.

WHAT IT ENCODES

Serum complement (C3, C4) is NORMAL in ANCA vasculitis because the disease is pauci-immune and not complement-consuming. This helps separate it from the LOW-complement nephritic GNs.

BOARD TRAP

The WRONG answer is LOW C3/C4 — that signals a complement-consuming/immune-complex GN: post-streptococcal GN, lupus nephritis, membranoproliferative GN, or cryoglobulinemia. Normal complement with a pulmonary-renal syndrome points to ANCA vasculitis (or anti-GBM), not these.

5. GPA = PR3 / c-ANCA

WHAT YOU SEE

The left worker table labelled GPA — WEGENER'S, with kidney meat, lung nodules, ENT/sinus icons and a PR3 c-ANCA tag — the granulomas + upper-airway props mark this as the PR3/c-ANCA triplet.

WHAT IT ENCODES

GPA (formerly Wegener's) is associated with c-ANCA / anti-PR3 (proteinase-3) in ~80–90% of cases and features NECROTIZING GRANULOMATOUS inflammation of the upper airway (sinusitis, saddle-nose deformity, otitis, subglottic stenosis), lower airway (nodules/cavities, hemoptysis), and kidney. It tends to relapse more than MPA. Induction is steroids plus rituximab or cyclophosphamide.

BOARD TRAP

The WRONG pairing is MPO / p-ANCA for GPA — GPA is the PR3 / c-ANCA disease with GRANULOMAS and prominent ENT involvement. Assigning p-ANCA or 'no granulomas' to GPA is the classic swap with MPA.

6. MPA = MPO / p-ANCA

WHAT YOU SEE

The middle table labelled MPA — MICROSCOPIC POLYANGIITIS with an MPO p-ANCA tag and a crossed-out NO GRANULOMAS / NO LUNG-SINUS sign — the missing granulomas mark this as the MPO/p-ANCA triplet.

WHAT IT ENCODES

MPA is associated with p-ANCA / anti-MPO (myeloperoxidase) and causes pauci-immune necrotizing small-vessel vasculitis (glomerulonephritis, pulmonary capillaritis/alveolar hemorrhage, palpable purpura, mononeuritis multiplex) WITHOUT granulomas and WITHOUT the destructive upper-airway/sinus disease of GPA.

BOARD TRAP

The WRONG feature is GRANULOMAS — MPA has NONE (that is GPA). Also avoid assigning nasopharyngeal/saddle-nose disease to MPA. The discriminator from GPA: MPO/p-ANCA, no granulomas, no ENT destruction.

7. EGPA = asthma + eosinophilia

WHAT YOU SEE

The right table labelled EGPA — CHURG-STRAUSS reading ASTHMA + ALLERGY + EOSINOPHILIA, with an inhaler and flowers — the asthma/allergy props are the EGPA fingerprint.

WHAT IT ENCODES

EGPA (Churg-Strauss) features late-onset ASTHMA, allergic rhinitis/nasal polyps, marked peripheral EOSINOPHILIA (>10%), eosinophilic tissue infiltration, mononeuritis multiplex, and cardiac involvement (a leading cause of mortality). Only ~40–60% are ANCA-positive, and when positive it is usually MPO/p-ANCA; ANCA-positive patients more often have GN/neuropathy while ANCA-negative patients more often have cardiac/lung disease. Mepolizumab (anti-IL-5) is used in addition to steroids.

BOARD TRAP

The trap is expecting EGPA to be reliably ANCA-positive — it is ANCA-negative in roughly half of cases, so a negative ANCA does NOT exclude it. The clinching combination is asthma + eosinophilia + vasculitis, not the serology alone.

8. Hemoptysis / pulmonary

WHAT YOU SEE

A pair of bleeding lungs coughing red, labelled HEMOPTYSIS — the blood-filled alveoli depict pulmonary capillaritis / diffuse alveolar hemorrhage.

WHAT IT ENCODES

Pulmonary capillaritis causes diffuse alveolar hemorrhage — hemoptysis, dropping hemoglobin, diffuse alveolar infiltrates on imaging, and a RISING DLCO from intra-alveolar blood — which together with the glomerulonephritis constitutes the pulmonary-renal syndrome. This is a life-threatening presentation requiring urgent immunosuppression and often plasmapheresis.

BOARD TRAP

The differential for pulmonary-renal syndrome includes anti-GBM (Goodpasture) and lupus, not just ANCA vasculitis — the trap is assuming hemoptysis-plus-GN is automatically ANCA; serology (ANCA vs anti-GBM vs ANA/complement) and biopsy IF distinguish them.

9. High mortality if untreated

WHAT YOU SEE

A bright red warning sign reading HIGH MORTALITY UNTREATED — URGENT TX — the alarm-red placard stands for the lethal course without prompt immunosuppression.

WHAT IT ENCODES

Untreated ANCA-associated GN historically carried very high mortality and rapid progression to ESRD; outcomes improved dramatically with immunosuppression. Treat empirically once the diagnosis is strongly suspected rather than waiting for full confirmation, because every day of delay costs nephrons.

BOARD TRAP

The trap is withholding therapy pending every confirmatory result — in suspected RPGN with active disease, empiric high-dose steroids are started immediately while biopsy/serology are pending; delay leads to irreversible renal loss and death.

10. Plasmapheresis (severe)

WHAT YOU SEE

A PLASMAPHERESIS tanker truck hauling away plasma — the truck physically carting fluid off IS the plasma exchange that removes circulating ANCA in severe disease.

WHAT IT ENCODES

Plasma exchange removes circulating ANCA and inflammatory mediators and is reserved for SEVERE disease — notably diffuse alveolar hemorrhage and severe renal failure (e.g., dialysis-dependent or rapidly rising creatinine). After the PEXIVAS trial its routine use was scaled back, but it remains indicated for alveolar hemorrhage and concomitant anti-GBM disease.

BOARD TRAP

The trap is offering plasmapheresis as routine first-line for ALL ANCA vasculitis — it is NOT; it is reserved for severe presentations (alveolar hemorrhage, severe/dialysis-requiring renal failure, or overlap with anti-GBM). Standard induction is steroids + rituximab/cyclophosphamide.

11. Induction: steroids + cyclo/rituximab

WHAT YOU SEE

An INDUCTION shelf/wall holding steroid, cyclophosphamide and rituximab agents — the labelled trio is the induction regimen.

WHAT IT ENCODES

Induction therapy is high-dose glucocorticoids (often IV methylprednisolone pulses then oral prednisone taper) PLUS either rituximab (anti-CD20; preferred for relapsing disease, PR3/GPA, and to spare fertility) or cyclophosphamide. Avacopan (C5a-receptor antagonist) is a newer steroid-sparing add-on. Add PJP prophylaxis (TMP-SMX) given the immunosuppression.

BOARD TRAP

The trap is using steroids ALONE — monotherapy is inadequate for organ-threatening ANCA vasculitis; a steroid-sparing agent (rituximab or cyclophosphamide) is required for induction. Forgetting PJP prophylaxis is a second common oversight.

12. Maintenance 1–2 yrs

WHAT YOU SEE

A MAINTENANCE 1-2 YRS shelf of pill bottles — the long row of drugs depicts years of ongoing maintenance immunosuppression.

WHAT IT ENCODES

After remission, maintenance immunosuppression continues for at least 1–2 years (often longer in relapsing/PR3 disease) using rituximab (preferred) or azathioprine/methotrexate, with the steroid tapered off. ANCA titer trends and clinical surveillance guide relapse detection.

BOARD TRAP

The trap is stopping immunosuppression once the patient is in remission — relapse risk is high (especially GPA/PR3), so maintenance therapy and monitoring are continued for years, not discontinued at first remission.

13. Biopsy / ANCA titer

WHAT YOU SEE

A banner reading ANCA TITER MONITOR with relapse arrows — the dial-watching scene stands for tracking ANCA titers (plus biopsy) to follow disease activity.

WHAT IT ENCODES

Diagnosis rests on ANCA serology (PR3/c-ANCA vs MPO/p-ANCA) PLUS tissue confirmation — renal biopsy showing pauci-immune necrotizing crescentic GN. ANCA titers can track disease activity and relapse, though they should be interpreted alongside clinical findings rather than treated in isolation.

BOARD TRAP

The trap is diagnosing or re-treating on ANCA titer alone — a rising titer in an asymptomatic patient does not by itself mandate therapy, and a positive ANCA without compatible clinical/biopsy findings is not sufficient. Tissue biopsy remains the gold standard for confirmation.